

Computer Network
LAB-01

Ajanta

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Date _____

Experiment Title: Study of LAN Topology, Network Devices and Address configuration.

Problem Statement

Objective: To study different lan topologies, identify network devices, analyze connection types, measure network speed and obtain system network details.

Requirements: computer
Internet connection
LAN Cable
Command prompt

A Local Area Network (LAN) connects devices within a limited area such as a college or office. Network topology defines how devices are arranged and connected.

Common topologies: Star Topology
Bus
Mesh

Procedure:

- (1) Draw the network Topology:
Hybrid Hierarchical Topology which combines
① Tree Topology

- ② Extended star topology
- ③ Ring topology (server room)

The network hierarchical layer:-

- ① Core layer
- ② Distribution
- ③ Access layer

(i) Server Room mesh Topology:

All switches are interconnected using fiber uplink core switch -1 and core switch -2 directly connected to each other and to both Distribution switch -1 and Distribution switch -2. Router/Firewall, Server, and Network monitoring PC connected to core switches using Ethernet LAN. This forms the central core of the campus.

(ii) Academics 1st, 2nd, 3rd floor - star topology:

Each floor connects to the server room via fiber uplink from core switches to the floor switches. Within each floor switch -1 and switch -2 connect to Access points AP-1 to AP-6 using Ethernet LAN or wifi. Each AP has a dedicated connection to the central switch.

(iii) Girls Hostel - Bus topology:

Hostel Switch connects to server room via Fiber uplink. Aps tap into the common bus cable.

(iv) Boys Hostel - Ring topology:

APs connected in a closed circular loop. Each AP links to two neighbors.

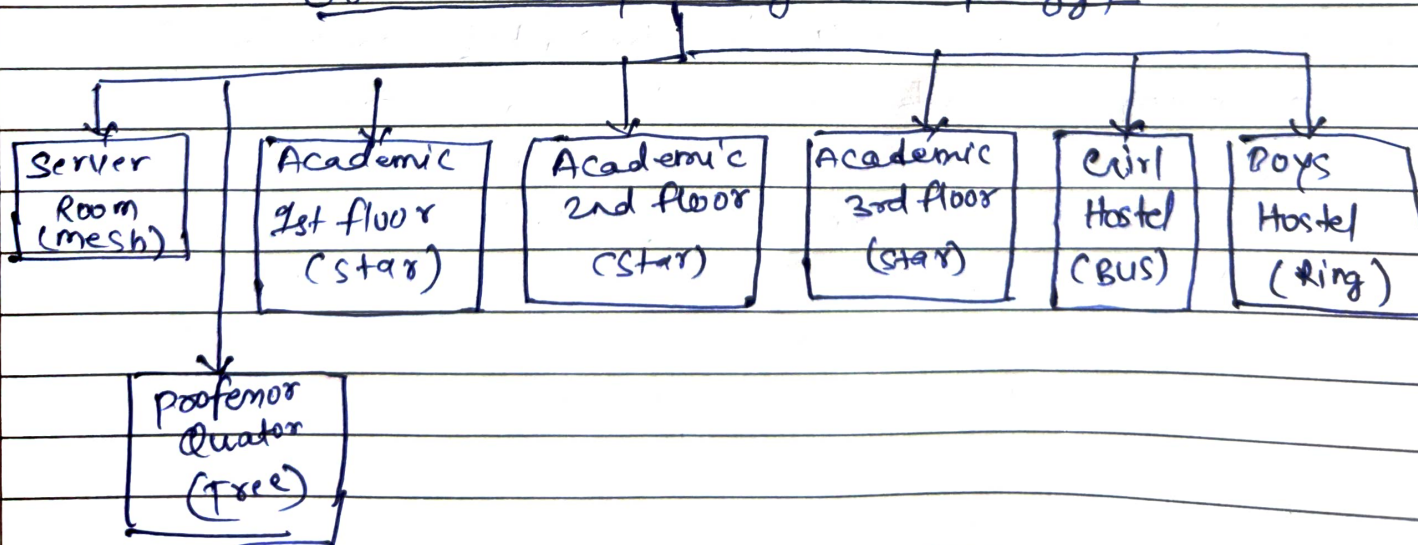
(v) Professor Quater - Tree Topology:

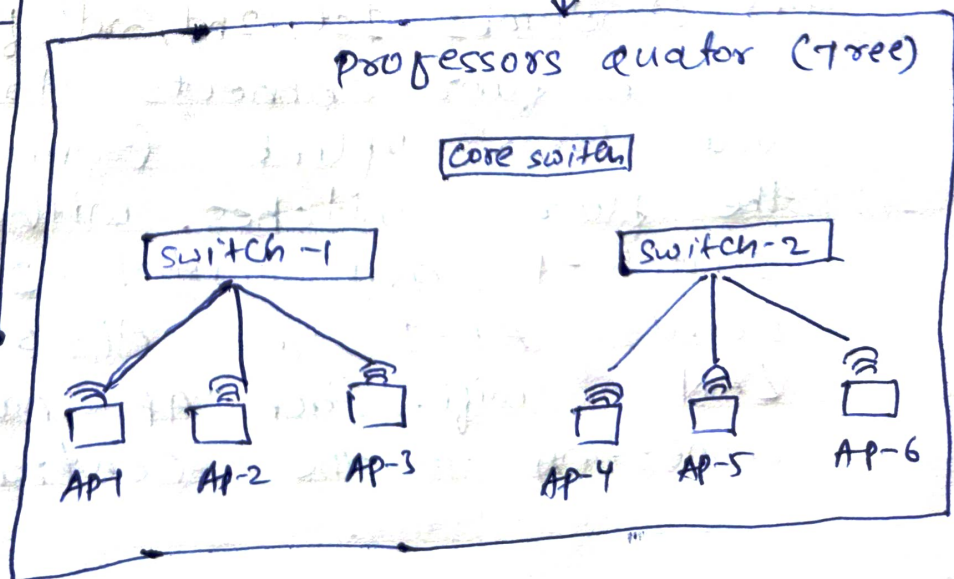
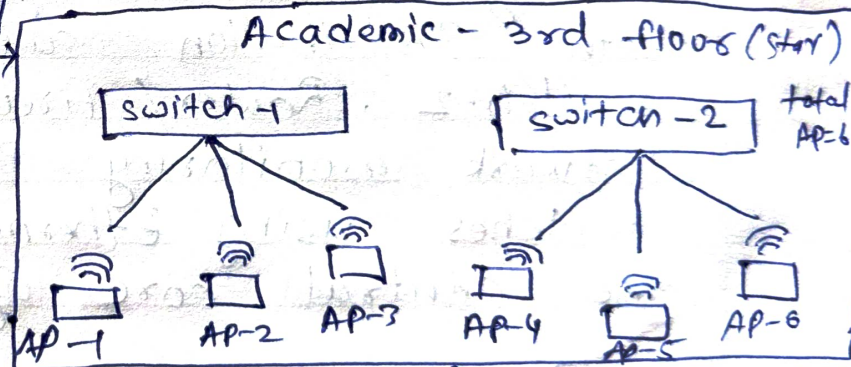
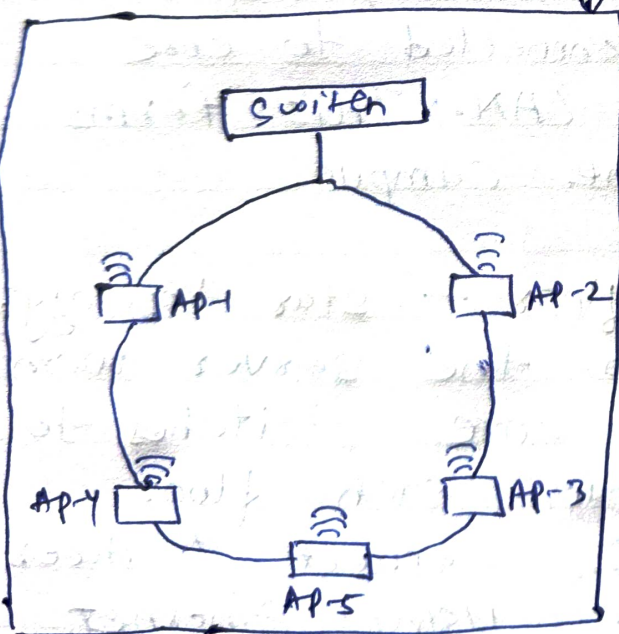
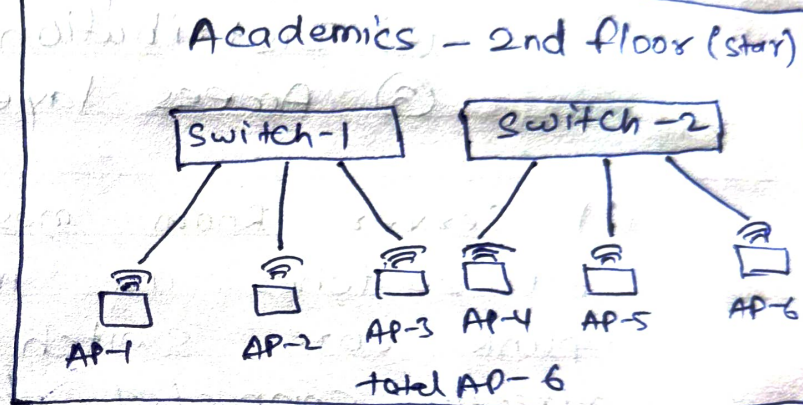
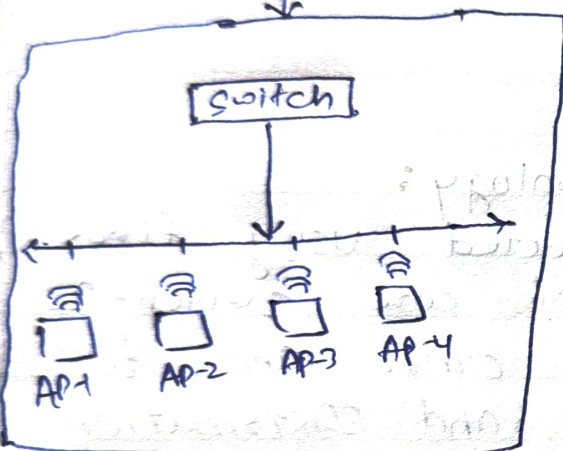
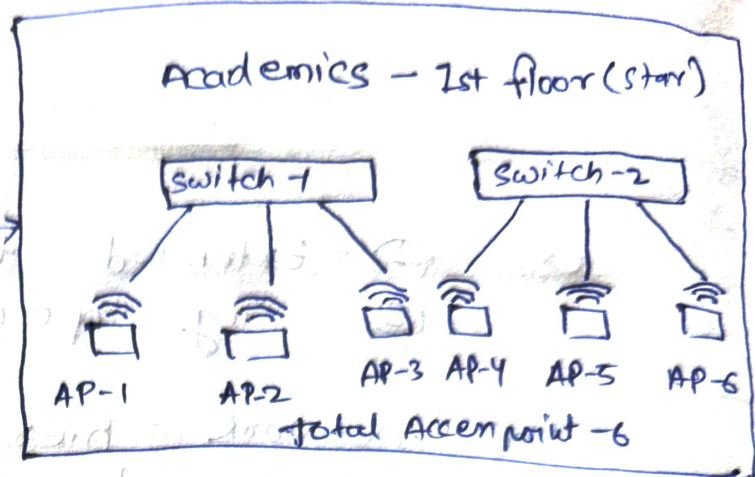
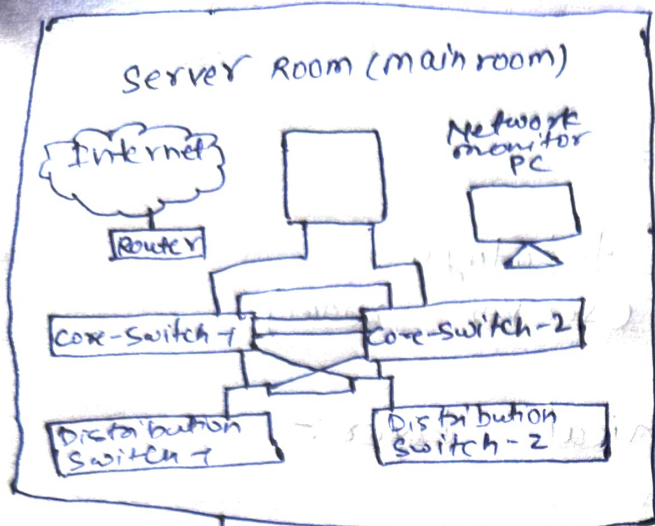
Core Switch connects to server room via fiber uplink. Sub-switches connect to APs via Ethernet.

(vi) Overall campus - Hybrid Topology:

Mix of Mesh, Star, Bus, Ring, Tree in one Network.

Overall campus (Hybrid Topology):





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Command Prompt
C:\Users\USER_1>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . . :

Ethernet adapter Ethernet 3:

    Connection-specific DNS Suffix . . :
    Link-local IPv6 Address . . . . . : fe80::fdbe:3046:c71a:6eb%10
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :

Wireless LAN adapter Local Area Connection* 9:

    Media State . . . . . : Media disconnected
    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . . :

Wireless LAN adapter WiFi:

    Connection-specific DNS Suffix . . :
    IPv6 Address. . . . . : 2401:4900:d16b:a944:a1bb:5713:b88e:3427
    Temporary IPv6 Address. . . . . : 2401:4900:d16b:a944:1c51:9961:b452:8b31
    Link-local IPv6 Address . . . . . : fe80::1e23:f1:429d:8b90%15
    IPv4 Address. . . . . : 10.83.223.229
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::6ca7:2aff:fe97:f345%15
                                10.83.223.133

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . . :

C:\Users\USER_1>
```

28°C
Mostly cloudy

08:59 AM
23-09-2026

- Processes
- Performance
- App history
- Startup apps
- Users
- Details
- Services

Performance



CPU
1% 3.33 GHz



Memory
8.6/31.4 GB (27%)



Disk 0
SSD (NVMe)
0%



Disk 1 (C:)
SSD (NVMe)
1%



Ethernet
Ethernet 3
S: 0 R: 0 Kbps



WiFi
WiFi
S: 0 R: 0 Kbps



GPU 0
NVIDIA GeForce RTX...
0% (37 °C)



GPU 1
AMD Radeon(TM) Gr...
0% (39 °C)

WiFi

Throughput

MediaTek Wi-Fi 6 MT7921 Wireless LAN Card



Send
0 Kbps

Receive
0 Kbps

Adapter name: WiFi

SSID: umbrell@

Connection type: 802.11ac

IPv4 address: 10.83.223.229

IPv6 address: 2401:4900:d16b:a944:a1bb:5713:b88e:3427

Signal strength: 